

Adaptive Continuous Authentication

Prototype Stage Report

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1 Introduction

The current prototype connects keyboard activity to a trust score on a web dashboard. This report records the modules completed, the checks performed and the work remaining at the prototype stage.

2 Methodology and Design

Implemented modules

Module	Current implementation
Login and interface	Flask login page, typing area, trust graph and decision history.
Typing measurements	Key timings, typing speed and pauses grouped into 10-second windows.
Model and scoring	Saved Isolation Forest model connected to a 0–100 trust score and recommendation.
Recorded replay	Stored feature windows can be played through the same scoring engine.

Integration completed

Live typing and recorded replay provide two ways to demonstrate the prototype. Both reach the same scoring engine and display their results on the dashboard.

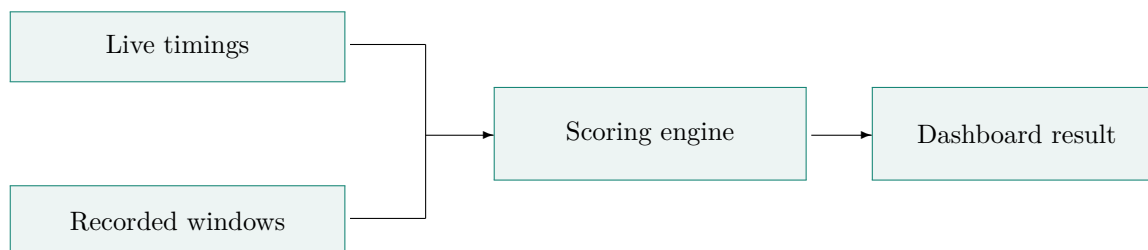


Figure 1: The two implemented input paths share one scoring engine.

Live capture is limited to the dashboard typing area. The browser sends timing information and coded key identifiers without sending the typed text.

3 Results and Evaluation

The repository’s automated checks cover login, typing measurements, replay, idle input and session handling. The previous local validation completed with all 11 tests passing.

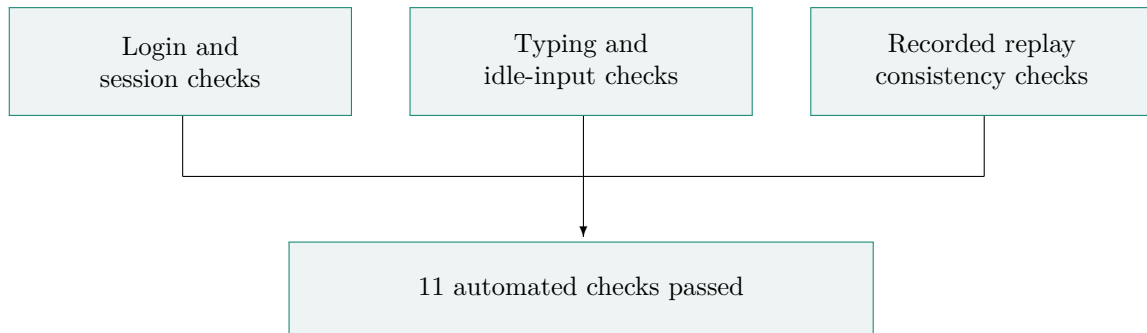


Figure 2: Software checks used to review the prototype.

Observed outcomes

Check	Observation
Recorded playback	All seven stored windows produced results consistent with the backend.
Typing measurements	Controlled input produced the expected timing values.
Inactive input	The previous trust score and recommendation were preserved.
Session handling	Login, invalid inputs, separate sessions and logout were checked.

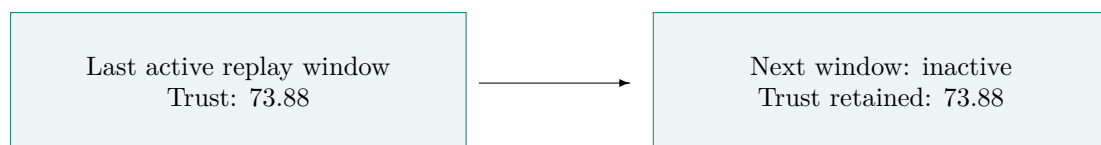


Figure 3: One observed replay result: inactivity keeps the previous score.

These checks establish the software’s behavior. The model uses only six active training windows, so authentication accuracy has not been established. The displayed score is not a probability of user identity.

4 Conclusion and Pending Work

The completed result is a working demonstration of typing input, scoring and dashboard feedback. Re-authentication and locking are recommendations only. More recording sessions, evaluation on unseen sessions and actual response enforcement remain pending.